higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

PROCESS CONTROL
NQF LEVEL 4

(6050414)

19 November 2019 (X-Paper)
09:00–12:00

This question paper consists of 4 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each section on a NEW page.
5. Use only a BLUE or BLACK pen.
6. Write neatly and legibly.

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QUESTION 1

1.1 Define the following terms:

1.1.1 Variable data

1.1.2 Non-conformance

1.1.3 Quality

1.1.4 Statistical process control

(4 × 2) (8)

1.2 Describe the THREE steps that are used in the manufacturing process to perform quality checks. (3)

1.3 A control chart is a graphical tool used by quality technicians to control and analyse the production process.

Name FIVE uses of control charts. (5)

1.4 Explain TWO problems that may occur if data is not properly collected. (2)

1.5 What does the acronym OHS stand for? (2)

1.6 List FIVE instruments used to measure differential pressure. (5)

[25]

QUESTION 2

2.1 Name THREE types of sampling processes. (3)

2.2 Name TWO disadvantages of in-line analysers. (2)

2.3 Name THREE types of control systems. (3)

2.4 Briefly explain when it is important to use manual control. (5)

2.5 Plant shutdown is one of the major activities in process plants.

Briefly explain the reasons for a plant shutdown. (6)

2.6 Draw a flow diagram that shows a logic control set up. (6)

[25]

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QUESTION 3

3.1 Define the term *automation system.* (2)

3.2 Name TWO functions of automatic control. (2)

3.3 List THREE forms in which signals can be converted by transducers. (3)

3.4 Write brief notes about remote terminal units. (4)

3.5 List SIX reasons for a data retrival system. (6)

3.6 Draw the ISO PFD symbols for the items below:

3.6.1 Filter

3.6.2 Shell and tube heat exchanger

(2 × 3) (6)

3.7 Name TWO methods of data capturing. (2)

[25]

QUESTION 4

4.1 List FIVE different forms of data retrival. (5)

4.2 For data to be quickly and effectively retrieved, data retrieval screens are used as part of the DCS system.

Briefly explain how data retrival screens operate. (5)

4.3 Draw a basic control chart showing all its components. (5)

4.4 Define the term *abnormal deviations.* (2)

4.5 Explain how the following Ms can cause process deviations:

4.5.1 Material

4.5.2 Machines

(2 × 2) (4)

4.6 Various safety devices can be used to protect people from exposure to hazards during start-up by preventing machines from starting.

Describe FOUR safety devices used during start-up. (4)

[25]

TOTAL: 100

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